

Ischemic CVA — Work-up & Management Reference

5-step spine: r/o bleed → reperfuse w/o ICH → etiology → secondary ppx → risk modification. Adapted from TeachIM (Fainstad, 2020) · 2021 + 2024 AHA/ASA · CC BY-NC 4.0.

Step 1 — Exclude hemorrhage

- Emergent **noncontrast head CT** first — ~85% ischemic, ~15% hemorrhagic
- No antithrombotic / no lytic until ICH excluded

Step 2 — Reperfuse without ICH

Therapy	Trigger & action
IV lysis	LKW <4.5 h, no CI, BP <185/110 → alteplase 0.9 or TNK 0.25 mg/kg (2024 Class 1). Door-to-needle <30 min
EVT	CTA LVO + salvageable brain (ASPECTS ≥6 or CTP mismatch); <6 h, sel to 24 h. Door-to-groin <90 min
Large-core EVT	Ant LVO, 6–24 h, NIHSS ≥6, ASPECTS 3–5 , age <80 (SELECT2/ANGEL-ASPECT 2023)

Contraindications to lysis: prior ICH, surgery <3 mo, active bleed, coagulopathy, BP >185/110.

Acute BP: permissive <220/120 ×48 h; **<185/110 if lysed/EVT**. NPO til dysphagia screen; SCD DVT ppx (pharm held <24 h post-tPA); avoid lines/tubes; PT/OT/SLP.

Step 3 — Determine etiology

Standard: MRI brain, H&N CTA/MRA, carotid/vert/TCD, telemetry, TTE-bubble.

Imaging pattern	Etiology
Large / multi-vessel territory	Cardioembolic
Lacunar (small, deep)	HTN / amyloid
Small/single + risk factors	Atherosclerosis

- TEE** only if mod-high cardioembolic suspicion w/ no TTE source or confirmed shunt
- Thrombophilia** testing: defer, heme/neuro-guided, mod-high pretest prob only

Step 4 — Secondary prophylaxis

Etiology	Antithrombotic
Atherosclerotic	Antiplt; if NIHSS ≤3 or ABCD2 ≥4 → DAPT (clopidogrel+ASA) ×21 d then mono (CHANCE/POINT). Ticagrelor+ASA ×30 d alt (THALES)
Cardioembolic	Anticoagulation after hemorrhagic-conversion window (antiplt bridge ~1–2 wk for large infarct)
Cryptogenic + PFO <60 y	Consider PFO closure (RESPECT/CLOSE/REDUCE)

Step 5 — Risk modification

- High-intensity statin to **LDL <70** (± ezetimibe/PCSK9)
- BP <130/80 long-term; A1c <7; glucose <180 acute inpatient
- Tobacco cessation; diet (limit processed foods)

Do / AVOID

DO	AVOID
Give lytic AND pursue EVT (complementary)	Withholding EVT for "large/late" stroke without imaging selection
Permissive HTN if not reperfused	Over-lowering BP (<140/90) in non-reperfused stroke
Delay full anticoag in large cardioembolic infarct	Early full anticoagulation → hemorrhagic conversion
Match antithrombotic to etiology	Routine acute thrombophilia panel